



United International University

School of Science and Engineering

Department of Computer Science and Engineering

Mid Term Exam

Trimester:- Spring 2026

Course Title: Fundamental Calculus

Course Code: MATH-1151

Marks:- 30

Time: 1 hour 30 minutes

[Note that the marks are given in brackets at the end of each question or part question. You have to answer all the questions in order.]

1. (a) The graph of a piecewise function $f(x)$ is given as follows:

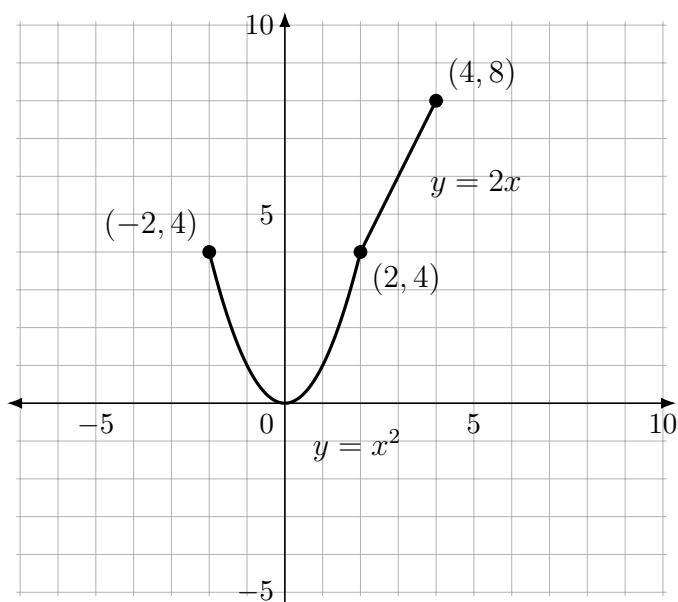


Figure 1

- i) Write down the function $f(x)$. [2]
 - ii) What is the gradient/slope of the 2nd piece? [1]
 - iii) Find the domain and range of the function. [2]
 - iv) What is the minimum value of the function? [1]
 - v) State whether the function is continuous at $x = 2$ or not. [1]
- (b) The graph of a function defined for $x \geq 0$ is given as follows: [2]

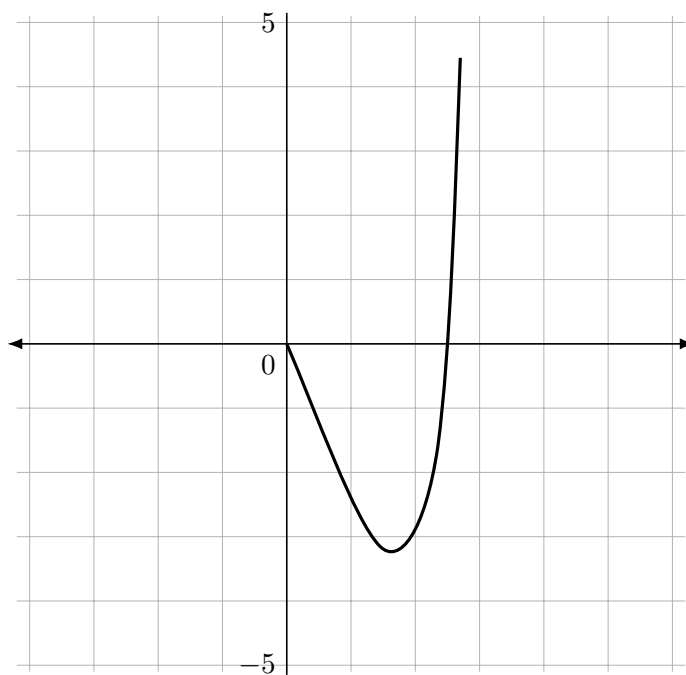


Figure 2

Complete the graph for $x < 0$ to make, (i) an even function and (ii) an odd function.

2. (a) Consider the function, $f(x) = \sqrt{2x - 5}$.
 - i) Sketch $f(x)$. Hence, determine whether the function $f(x)$ is one-one or not. [2]
 - ii) Find the inverse $f^{-1}(x)$ of $f(x)$. [2]
 - iii) What is the range of $f^{-1}(x)$? [1]
 - iv) Draw the graph of $f(x)$ and $f^{-1}(x)$ on the same diagram. [2]
- (b) Given that, $g(x) = 7x^2 + 35$. Find $g \circ f$. [2]
3. (a) The graph of the function $f(x)$ is given as follows:

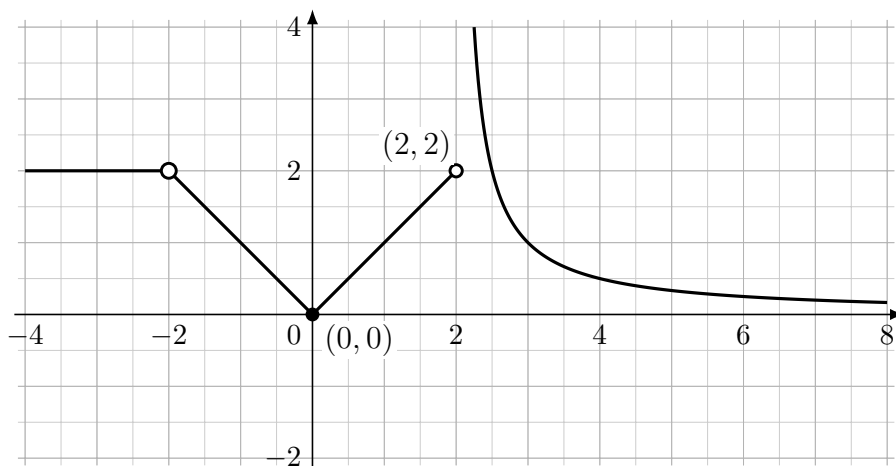


Figure 3

State the followings:

i) $\lim_{x \rightarrow \infty} f(x)$ [1]

ii) $\lim_{x \rightarrow 2} f(x)$ [1]

iii) $\lim_{x \rightarrow -2} f(x)$ [1]

(b) Given that, $f(x) = 7 + 9 \ln(1 + \frac{1}{x})$. Find the value of $\lim_{x \rightarrow \infty} f(x)$. [1]

(c) Given that, $f(x) = \frac{3x+2}{x-3}$. Find Vertical and Horizontal asymptotes.
Hence, sketch $f(x)$. [2]

4. (a) Describe in words the transformations,

i) $y = 2f(x) + 1$. [2]

ii) $y = f(1 - x)$ [2]

(b) The graph of the function $f(x)$ is given as follows:

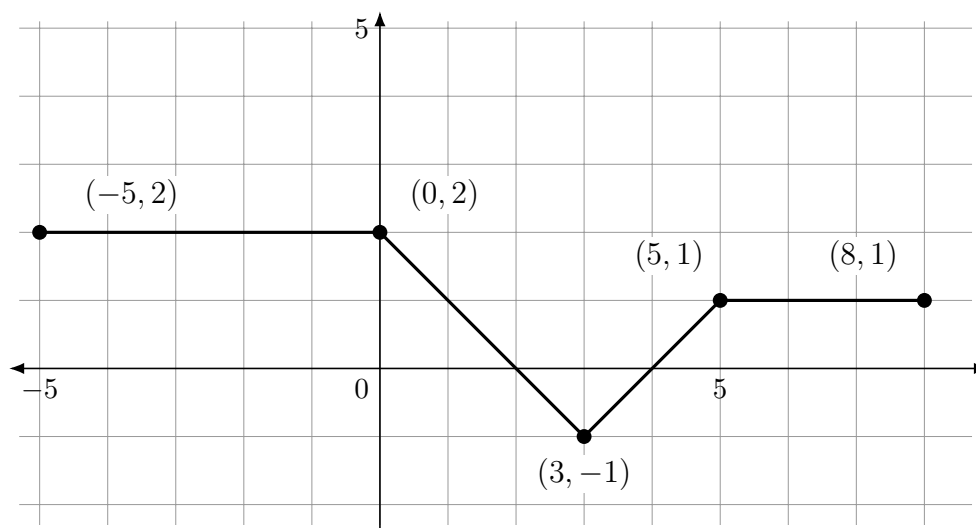


Figure 4

Sketch $y = |-f(x)|$. [2]